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TOPICAL REVIEW

Optimal Power Flow: A Review of State-of-the-Art Techniques and Future Perspectives

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ABSTRACT The Optimal Power Flow (OPF) problem has become increasingly pivotal in the planning and operation of modern power systems. With the expansion of the grid scale, the advent of smart grid technologies, and the unpredictable nature of renewable energy sources (RESs), interest in OPF has surged. These challenges with new energy storage have introduced a heightened level of uncertainty into the power system's operation as well as planning. Because of this, OPF is seen as an important tool for achieving different goals, such as optimizing the distribution of resources, making electrical networks more efficient, and so on. However, the OPF problem is inherently difficult to solve because of its non-linear characteristics. Different constraints and limitations intrinsic to real power system grids further accentuate this complexity. Moreover, modern power systems have incorporated new constraints, which make the OPF problem more complex in terms of mathematical formulation and solution. This paper offers a comprehensive and foundational review of OPF, covering the main concept, mathematical formulation, OPF types, comprehensive OPF optimization problem concepts, and the various methods developed to solve it. Additionally, it explores the evolution of these methods from conventional approaches to advanced and recent techniques, including mathematical methods and artificial intelligence methods, which include metaheuristic (search-based) and machine learning algorithms (data-driven). The paper also discusses various types of convex relaxation methods in depth. Ultimately, the paper highlights key gaps, challenges, and opportunities for future research.

INDEX TERMS Optimal power flow (OPF), machine learning, metaheuristic optimization algorithms, OPF convex relaxation, artificial intelligence methods, renewable energy systems, economic dispatch formulation.

A. ABBREVIATIONS

NOMENCLATURE

ABC Artificial Bee Colony.
 ACO Ant Colony Optimization.

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BA Barnacle Algorithm.
 BBO Biogeography-Based Optimization.
 BESS Battery Energy Storage Systems.
 BFA Bacterial Foraging Algorithm.
 BHA Black Hole Algorithm.
 BMO Barnacles Mating Optimization.

BSA	Backtracking Search Algorithm.
CNN	Convolutional Neural Networks.
CS	Cuckoo Search.
DF	Differential Evolution.
DGs	Distributed Generators.
DNN	Deep Neural Networks.
ED	Economic Dispatch.
EO	Equilibrium Optimizer.
FACTS	Flexible Alternating Current Transmission Systems.
FDA	Flow Direction Algorithm.
GA	Genetic Algorithm.
GOA	Grasshopper Optimization Algorithm.
GRO	Group Optimization.
GSA	Gravitational Search Algorithm.
GSO	Glowworm Swarm Optimization.
GWO	Grey Wolf Optimizer.
HAS	Harmony Search.
IMPs	Interior Point Methods.
IWDO	Improved Wind-Driven Optimization.
LP	Linear Programming.
MBA	Migrating Birds Optimization Algorithm.
MFA	Multi-Verse Optimizer.
ML	Machine Learning.
MSA	Moth Search Algorithm.
NN	Neural Network.
NR	Newton-Raphson.
PF	Power Flow.
PINN	Physics-Informed Neural Network.
PSO	Particle Swarm Optimization.
QC	Quadratic Convex.
QP	Quadratic Programming.
RESs	Renewable Energy Sources.
SA	Simulated Annealing.
SCO	Sine Cosine Optimization.
SCOPF	Security-Constrained Optimal Power Flow.
SDP	Semidefinite Programming.
SOCP	Second-Order Cone Programming Relaxation.
SOS	Symbiotic Organisms Search.
SSA	Salp Swarm Algorithm.
SVC	The Static VAR Compensator.
TCPS	Thyristor-Controlled Phase Shifter.
TCSC	Thyristor-Controlled Series Capacitor.
TLBO	Teaching-Learning-Based Optimization.
WOA	Whale Optimization Algorithm.

B. SYMBOLS

P_{Gi} :	Active power generation at bus i
P_{Di} :	Active power demand at bus i
V_i :	Voltage magnitude at bus i
V_j :	Voltage magnitude at bus j
ϕ_{ij} :	Voltage angle difference between bus i and bus j
G_{ij} :	Real part of the admittance of the transmission line between bus i and bus j

B_{ij} :	Imaginary part of the admittance of the transmission line between bus i and bus j
N_b :	Total number of buses in the system
Q_{Gi} :	Reactive power generation at bus i
Q_{Di} :	Reactive power demand at bus i
$\mathcal{G}$:	Set of all generators in the system
$\mathcal{B}$:	Set of all buses in the power system
$\mathcal{L}$:	Set of all transmission lines in the power system
$\mathcal{S}$:	Set of all buses where devices providing reactive power support
$\mathcal{T}$:	Set of all transformers in the power system
$\mathcal{W}_{ii}$:	Diagonal element of matrix W
$\mathcal{W}_{ij}$:	Off-diagonal element of matrix W
$\mathcal{W}_{n \times n}$:	Matrix W of size $n \times n$

I. INTRODUCTION

A. RESEARCH OBJECTIVE

In the past two decades, the complexity of electric power systems has grown significantly due to several key factors, such as increased power demand and the unpredictable generation from renewable energy sources (RESs), which play major roles. Furthermore, these systems' vast scale, numerous dynamic and static states, and complex component interfaces compound the challenge [1], [2], [3]. Optimal power flow (OPF) is considered one of the crucial tools in this process and serves a important role in the operation and control of power grids. Since the OPF concept was introduced by [4] in 1962, it has grown to be one of the most widely researched and essential fields in power system operation. It finds the best way to minimize or maximize objective functions while still meeting PF constraints and other operational limits, such as voltage limits, transmission stability, and minimum and maximum generator output requirements [5]. However, Due to the nonlinearity of PF equations and constraints [6], OPF problems are inherently non-convex and nonlinear [7]. This means that there are multiple local optima, which makes it very hard to find the global optimum and becomes a computational problem [8], [9]. The studies [10] and [11] gave a strong NP-hardness proof for the AC-OPF problem, which means that it is computationally impossible to find workable solutions in large power networks.

In their paper [12], Montoya et al. suggested a new repetitive method for dealing with the OPF in DC systems that can be used instead of the more common Gauss-Seidel and Newton-Raphson methods. The aforementioned methodologies bear resemblance to both the traditional Gauss-Seidel technique and the conventional Newton-Raphson technique. Findings suggest the developed approach exhibits superior efficacy regarding result accuracy and processing efficiency. The authors in [13] used Taylor's series development to find the best way for OPF in microgrids. They used nonlinear massive algorithms to compare the suggested linearization with the standard nonlinear nonconvex paradigm. The investigations in [14], [15], and [16] examined OPF research regarding

solutions and dynamic approaches, alongside unpredictable as well as mixed approaches. A recent thorough assessment of advancements and necessary additional enhancements in AC OPF has been documented in [17] and [18]. The existing OPF approaches are frequently articulated as predictable optimization challenges, which fail to account for unknown factors like as demand fluctuations and the unpredictability of clean energy resources [19], [20], [21], [22]. Consequently, the OPF issue has been converted into the stochastic OPF dilemma in many recent studies. The advancement of stochastic solution approaches and the execution of bayesian optimal power flow (B-OPF) studies have grown into essential prerequisites and valuable decision-making instruments in the design, functioning, and management of active distribution networks (ADNs) [23], [24], [25], [26]. Moreover, the P-OPF is recognized for its capacity to accomplish multiple goals such as but not restricted to, minimizing power losses, reducing overall generating costs, decreasing voltage deviations [27], [28], and lowering carbon emissions. This requires a precise model for all aspects of the system and an effective solution mechanism [29], [30].

OPF calculations are a crucial part of managing power systems every day, particularly for clearing electricity markets. These calculations need to be fast and reliable to keep up with the constantly changing demands and conditions of the power grid [31], [32]. Although many advanced OPF algorithms have been developed by researchers, system operators still rely heavily on the megawatt (MW)-only direct current (DC) OPF method for several decades [33], [34]. In the last few decades, the structure of the market energy sector has gradually changed to provide a higher contribution of renewable energy sources. While this conversion to renewables brings many advantages, including low cost of operation and fewer risks associated with pollution from fossil fuels and conventional resources, it creates a number of problems in the planning and operation of electricity networks [35], [36]. Nonetheless, the OPF problem is mathematically complex and nonlinear; it turns out to be more challenging when combined with conventional thermal power plants and fluctuating renewable resources such as wind and solar energy. Many recent studies have concentrated on solving the OPF problem when integrated with renewable energy [37], [38], [39], [40].

This paper provides a comprehensive and critical theoretical literature review of the OPF problem, including OPF formulation, types, and the methods employed to solve it to date. Moreover, it emphasizes the evolution of ACOPF methods from traditional optimization techniques to advanced metaheuristic methods, convex relaxation techniques, and machine learning (ML) approaches. It provides a critical review of metaheuristic algorithms in handling nonlinearity and nonconvexity in OPF problems, after which it explores convex relaxation techniques such as SDP and SOCP. Moreover, it elaborates on the use of machine learning approaches to address OPF specifically in terms of power

flow solution prediction, enhancing the traditional solvers. This paper points out many problems with current OPF methods and gives useful information about where future research should go. This work not only gives an honest assessment of current methods, but it also plans out important gaps and new directions for OPF research, showing how ACOPF could be improved.

The following sections of this paper are organized as follows: A comprehensive overview of the OPF formulation, including the different types of OPF, key optimization concepts, and the mathematical formulation provided in Section II. Section III discusses the conventional and mathematical methods, used to address OPF. Section IV reviews artificial methods, including both search-based and data-driven approaches. Section V covers the various convex relaxation methods. Several gaps, challenges, and future directions are explored in Section VI.

II. OPF FORMULATION

OPF combines the ideas of economic dispatch (ED) and PF analysis in order to guarantee that power systems operate optimally. PF tries to find a possible operating state by solving equations related to generation, load, and the transmission lines network. Often, they don't think about practical limits like generator reactive power limits or transmission line capacities. ED seeks to reduce the cost of power generation while satisfying load demand; however, the power flow limitations are ignored [41], [42], [43]. OPF uses both of these methods together to find the best answer to an objective function, like minimizing generating costs. It also takes into account operational constraints like transmission stability, generator output, and voltage limits, as well as power flow constraints. Additionally, OPF seeks to optimize a particular objective function, either minimizing or maximizing it, while satisfying both equality and inequality constraints. Various versions of OPF exist, including the following:

- Static OPF is a type that is aimed at identifying the optimal operating conditions for a power system at a specific time for a fixed load, which means It does not consider variations in load [44].
- Dynamic OPF determines the best operating points by optimizing the power system's operations over a continuous time frame, considering load changes [45], [46], [47].
- AC OPF is a type based on the natural inherent PF characteristics. AC OPF produces extremely exact findings by precisely modeling the nonlinear characteristics of AC systems. It seeks to discover the ideal operating conditions for an AC power system taking into account actual and reactive power flows, voltage levels, and phase angles [42], [48].
- DC OPF simplifies the optimum power flow problem by concentrating on a linear approximation of the power system. It assumes that voltage levels stay the same and that reactive power flows don't matter. This makes

it possible to linearize the complicated nonlinear AC power flow equations [49], [50].

- AC/DC OPF is a type of OPF that integrates characteristics and constraints of both DC and AC. It involves OPF for both AC and DC grids [51].
- The Transient Stability-Constrained (TSC-OPF) type of OPF maintains the stability and cost-effectiveness of power systems. The TSC-OPF uses both dynamic and steady-state constraints to keep the system stable during and after disturbances like faults or sudden changes in load [52]. In other words, it combines elements of both static and dynamic OPF [53], [54].
- Probabilistic OPF figures out the probability distribution functions of dependent variables by looking at the probability distributions of loads and other uncertainties [55] using the Point Estimate Method (PEM) [56], the Monte Carlo method [57], and a customized Gaussian mixture model [58].
- Stochastic OPF is a type of power system that manages and optimizes the power flow while accounting for uncertainty in various parameters. Compared to classic deterministic OPF, which assumes set values for system parameters, stochastic OPF takes into account unpredictability and randomness in inputs like load demands, renewable energy generation, and equipment reliability [59], [60], [61].
- Deterministic OPF is a form of OPF that assumes that all system parameters are known and fixed. It does not consider uncertainties or fluctuations [62] in inputs such as load demands and generating outputs [63].
- SCOPF considers both economic efficiency and system security. SCOPF accounts for potential contingencies, such as generator outages and transmission line failures [64], [65].

A. MATHEMATICAL OPF FORMULATION

This section outlines a comprehensive overview of the OPF problem formulation. OPF formulations vary greatly depending on the variables, objectives, and constraints used. The formulation changes have an impact on both algorithm design and solution accuracy. Variables are required in OPF issues to describe the electrical state of the power system. They are normally classified into control (independent) variables and state (dependent) variables. Control variables include active power generation at generator buses (with the exception of for the slack bus), voltage levels at generator buses, transformer tap settings, and reactive power injections from shunt capacitor banks. The variables are modified to accomplish the optimization objectives.

After identifying the optimal values for these control variables, state variables can be computed. State variables include the slack bus's active power output, voltage levels at load buses, generator reactive power generation, and transmission line loadings. The number of control variables determines the dimensionality of the solution space and

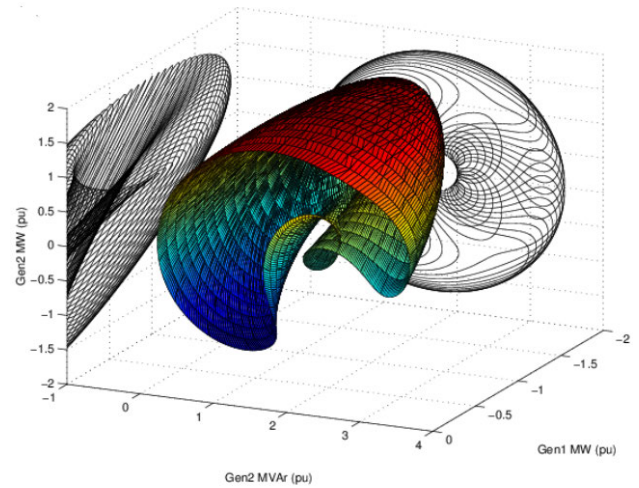


FIGURE 1. Feasible space of power flow [66].

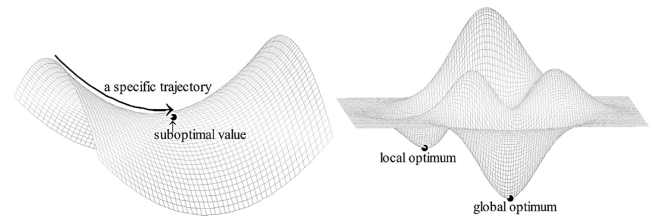


FIGURE 2. Local optimum and global optimum solutions [14].

hence influences the difficulty of the OPF problem. Setting these variables correctly is crucial for achieving objectives, for example, minimizing generating costs, decreasing power losses, and improving system reliability while adhering to constraints. FIGURE 1 illustrates a feasible space for power flow.

B. TYPES OF OPTIMIZATION SOLUTION

- A global optimal solution is the most optimal solution to an optimization problem, considering all potential solutions within the feasible set. It is the solution that yields the minimum (or maximum, depending on the problem) objective function value, ensuring that no other feasible solution results in a better outcome. In other words, a solution where the objective function obtains its minimum (or maximum) value across the entire feasible solution space, $f(x^*) \leq f(x)$ for all $x \in X$, where X is the feasible set.
- A local optimal solution is a solution that is the most favorable within a certain area of the search space, but it may not be the best solution overall. Every local optimum is also a global optimum in convex problems due to the characteristics of convex functions. In non-convex problems, on the other hand, there may be more than one optimal point. In these situations, a local optimum is a solution where there is no other optimal feasible solution nearby, but there may be better

solutions in other parts of the search space. In other words, a solution where the objective function attains its minimum (or maximum) value within a localized neighborhood of the feasible solution space, satisfying $f(x^*) \leq f(x)$ for all x in a neighborhood $N(x^*) \subseteq X$.

- A sub-optimal solution satisfies the essential requirements and improves on the initial conditions, but it is not the best solution. It is employed when obtaining the absolute best solution requires too much time and effort since it serves as a near approximation to the best solution. Figure 2 shows the different types of optimization solutions for non-linear problems.

C. OBJECTIVE FUNCTION IN OPF

The objective function is a mathematical expression or formula that finds the best way for a power system to work by lowering or raising certain criteria while still following all operational and physical limits. Common goals include minimizing the overall cost of power generation [67], decreasing transmission losses, reducing carbon emissions [68], lowering reserve costs, optimizing voltage profiles, and improving system security [69], [70]. Based on the goals, it can be approached with either single or multiple objectives [71], [72]. In a single-objective optimization, the goal is to either maximize or minimize a specific objective or to merge several objectives into one main goal. On the other hand, Multi-objective optimization seeks to accomplish several objectives at the same time such as minimizing generating costs, decreasing transmission losses, and enhancing voltage stability [73]. To achieve these goals, the OPF model adjusts key optimization variables such as generator active power output P_g , generator reactive power output Q_g , voltage magnitudes V , and voltage angles θ .

$$\min_{P_g, Q_g, V, \theta} f(P_g, Q_g, V, \theta) \quad (1)$$

1) FUEL COST MINIMIZATION

$$\text{Min } f = \sum_{i=1}^{N_G} f_i(P_{Gi}) = \sum_{i=1}^{N_G} (a_i P_{Gi}^2 + b_i P_{Gi} + c_i) \quad (2)$$

The valve-point loading effect (VLE) causes a ripple impact on the cost trajectory of thermal generators, resulting in an intensely non-convex expression. The modeling of fuel expenses using VLE modeled as:

$$\text{Min } f = \sum_{i=1}^{N_G} f_i(P_{Gi}) = \sum_{i=1}^{N_G} (a_i P_{Gi}^2 + b_i P_{Gi} + c_i) + |e_i \sin\{f_i \times (P_i^{\min} - P_i)\}| \quad (3)$$

The $|e_i \sin\{f_i \times (P_i^{\min} - P_i)\}|$ function in expression (3) represents the cost function with Valve-point Load Effect (VLE). As the power demand of a particular network increases, the connected generation units synchronize their output to meet the rising demand. This sudden increase in demand triggers a ripple effect in the cost curve, as turbine valves

are adjusted to increase the power output of the generators. The adjustment in valve positions to meet the heightened demand introduces a dynamic change in the system's cost structure, as the generation process becomes more responsive to fluctuations in demand.

2) POWER LOSS MINIMIZATION

$$\text{Min } f = \sum_{i,j \in N_b} g_{ij} (V_i^2 + V_j^2 - V_i V_j \cos \theta_{ij}) \quad (4)$$

3) VOLTAGE PROFILE IMPROVEMENT

$$\text{Min } f = \sum_{i=1}^{N_L} |V_{Li} - V^{\text{ref}}| \quad (5)$$

4) GENERATION EMISSION MINIMIZATION

$$\text{Min } f = \sum_{i=1}^{N_G} ((\alpha_i + \beta_i P_{Gi} + \gamma_i P_{Gi}^2) + |\zeta_i \exp[\lambda_i P_{Gi}]|) \quad (6)$$

D. CONSTRAINTS

Constraints are crucial for maintaining the safety and efficiency of the power system. These constraints may often be categorized into two types: equality constraints and inequality constraints.

1) EQUALITY CONSTRAINTS

$$P_{Gi} - P_{Di} - V_i \sum_{j=1}^{N_b} V_j (G_{ij} \cos \phi_{ij} + B_{ij} \sin \phi_{ij}) = 0 \quad (7)$$

$$Q_{Gi} - Q_{Di} - V_i \sum_{j=1}^{N_b} V_j (G_{ij} \sin \phi_{ij} - B_{ij} \cos \phi_{ij}) = 0 \quad (8)$$

2) INEQUALITY CONSTRAINTS

These limitations guarantee that the power outputs of generators stay within their designated limits and can be modeled as follows:

$$\begin{cases} P_{Gi}^{\min} \leq P_{Gi} \leq P_{Gi}^{\max}, \\ Q_{Gi}^{\min} \leq Q_{Gi} \leq Q_{Gi}^{\max}, \end{cases} \quad \forall i \in \mathcal{G} \quad (9)$$

3) VOLTAGE LIMITS

Voltage levels at all buses must be sustained within designated limits to guarantee system stability and dependability.

$$V_i^{\min} \leq V_i \leq V_i^{\max}, \quad \forall i \in \mathcal{B} \quad (10)$$

4) THERMAL LIMITS

The power flow on each transmission line must not beyond its thermal capability to avert overheating and damage.

$$P_{\text{line}}^{ij} \leq P_{\text{line}}^{ij, \max}, \quad \forall (i, j) \in \mathcal{L} \quad (11)$$

5) REACTIVE POWER SUPPORT

Securing sufficient reactive power assistance from generators and additional reactive power sources to sustain voltage stability.

$$Q_{\text{support},i}^{\min} \leq Q_{\text{support},i} \leq Q_{\text{support},i}^{\max}, \quad \forall i \in \mathcal{S} \quad (12)$$

6) FACTS DEVICES

Utilizing FACTS devices to enhance transmission capacity facilitates more efficient operation of the power system at reduced costs, while satisfying transmission and security standards [74]. Incorporating FACTS devices into the OPF necessitates the modification of the traditional OPF program by integrating control variables pertinent to these devices and altering the load flow solutions to tackle the associated issues. FACTS devices can swiftly and effectively regulate active and reactive power while concurrently modifying voltage magnitude. Regulating power flows throughout the network, under both standard and atypical situations, mitigates congestion on overburdened lines and reduces system power losses [75].

Primary problems include optimizing the use of existing infrastructure to ensure system security amid increased power transfers, which are limited by stability limits on transmission lines [76]. The enhancement of control techniques for FACTS devices and the delineation of performance requirements for their components. The operational properties of FACTS devices fluctuate according to the electricity system in which they are deployed. Many current algorithms are tailored for certain scenarios, hence constraining flexibility in establishing objectives and limitations. The integration of a TCSC device with a transmission line modifies its reactance which is given as:

$$X_{\text{new},ij} = X_{\text{line},ij} + X_{\text{TCSC},ij}, \quad \forall (i,j) \in \mathcal{L} \quad (13)$$

TCPS governs the phase angle of the voltage along a transmission line. The phase shift ϕ imposed by the TCPS is restricted to the following bounds:

$$\phi_{ij}^{\min} \leq \phi_{ij} \leq \phi_{ij}^{\max}, \quad \forall (i,j) \in \mathcal{L} \quad (14)$$

SVC regulates the injection or absorption of reactive power. The functioning of the SVC is regulated by the subsequent constraint:

$$Q_{\text{SVC},i}^{\min} \leq Q_{\text{SVC},i} \leq Q_{\text{SVC},i}^{\max}, \quad \forall i \in \mathcal{S} \quad (15)$$

Power flow equations, serving as equality constraints, must account for the impact of FACTS devices, whilst inequality constraints should ensure that the devices operate within their specified limits.

7) TRANSFORMER TAP SETTING

Through manipulation of taps, transformers are capable of controlling the voltage magnitude and phase and thus improving power flow direction and reducing losses in a system. The addition of tap-changing transformers to OPF not only improves the voltage profile but also helps control

a number of constraints, such as stability margins and thermal limits. The optimization of tap settings enables OPF with tap-changing transformers to operate securely and cost-effectively while adapting to dynamic grid conditions.

The transformer tap position should be within the range specified by the constraint:

$$T_{\min,i} \leq T_i \leq T_{\max,i}, \quad \forall i \in \mathcal{T} \quad (16)$$

Variables are necessary in OPF problems to characterize the electrical condition of the power grid. They are often divided into control (independent) and state (dependent) variables. Control variables involve active power generation at generator buses (except the slack bus), voltage levels at generator buses, transformer tap settings, and reactive power injections from shunt capacitor banks. The variables are modified to accomplish the optimization objectives. After identifying the optimal values for these control variables, state variables can be computed. encompass the active power output of the slack bus, voltage levels at load buses, generator reactive power generation, and transmission line loads. The quantity of control variables dictates the dimensionality of the solution space and hence influences the difficulty of the OPF problem. Setting these variables correctly is crucial for achieving objectives, for example, minimizing generating costs, decreasing power losses, and improving system reliability while adhering to constraints. Figure 3 illustrates the methods used to solve the OPF problem in this study.

III. CONVENTIONAL METHODS

Conventional or deterministic optimization methods have long been regarded as the principal solution for addressing OPF problems. These strategies utilize mathematical programming to enhance the performance of power systems while adhering to certain operational and technical restrictions. The number of recent papers published in several databases over the past few years is represented in Figure 4. As can be seen, it has steadily increased over the years verifying the importance of OPF.

A. LINEAR PROGRAMMING

The objective function and constraints are naturally not linear, but LP methods solve the OPF problem by turning them into a linear form. Consequently, the LP method efficiently solves the OPF problems through the linearization process. In the study [77], an extended OPF algorithm based on successive linear programming is introduced. This algorithm uses second-order sensitivity approximations to make the method more accurate and faster. This approach is particularly beneficial for systems where generation cost functions are similar, which typically causes slow convergence in traditional methods. [78] presented a penalty function-linear programming (PF-LP) approach to solve constrained ED (CED) problems in power systems. The approach integrates penalty functions into the LP framework to handle various constraints efficiently. The penalty function approach might

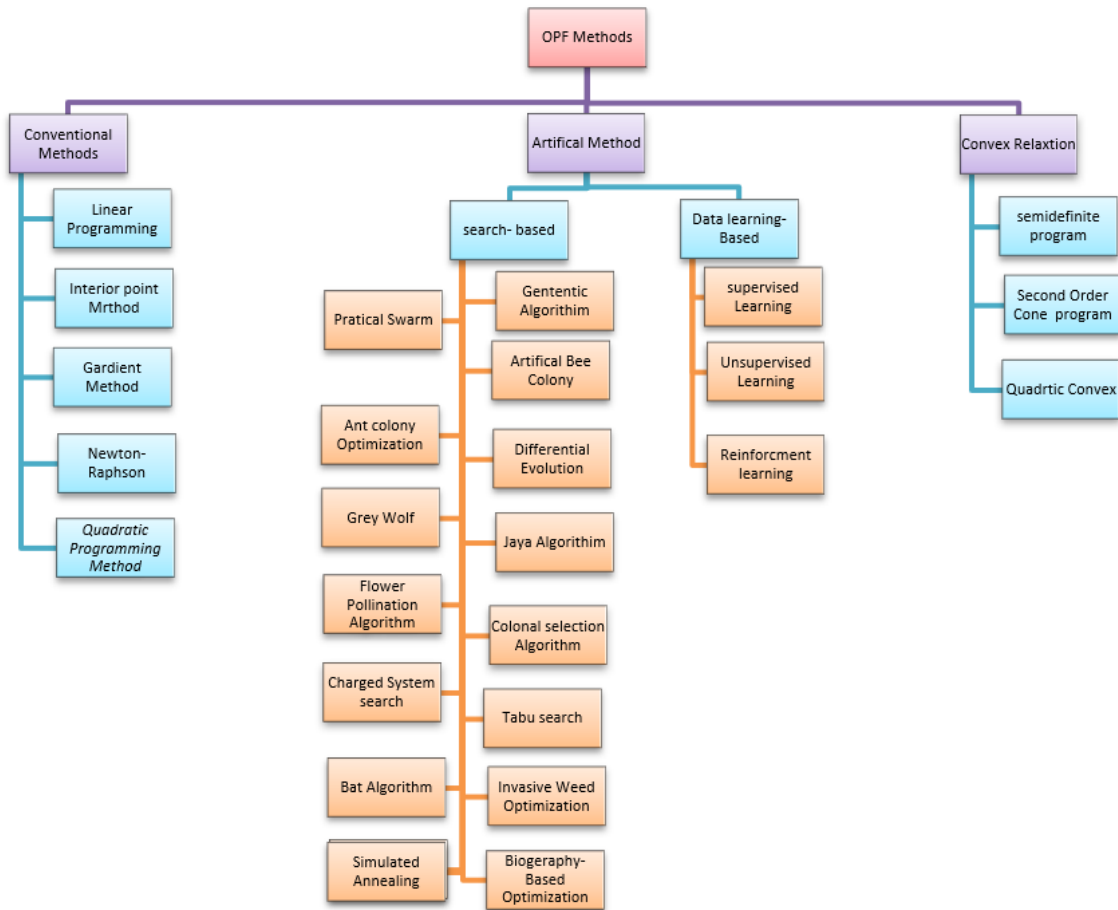


FIGURE 3. OPF methods discussed in this study.

not be able to handle impossible situations because it usually relies on an initial point that can be used for the LP solution process.

In cases where the problem constraints are infeasible at the initial point, the penalty function method may not be directly applicable, potentially restricting its effectiveness. A piecewise differentiable penalty function is used to tackle the constrained economic operation problem [79]. This technique is designed to manage CED with linear constraints, and it has proven useful not only in dual linear programming formulations but also in quadratic programming problems. [80] showed how to improve the solution by solving a series of linear LP problems over and over again while approximating the nonlinear objective function and constraints with linear functions. The method may require numerous iterations for highly accurate results, leading to increased computational effort. Additionally, the effectiveness of the SLP method can be challenged by highly nonlinear constraints, which might affect the approximation quality.

B. INTERIOR POINT METHOD

Interior Point Methods, introduced in 1984 ([81]), comprise a set of algorithms that employ projective scaling techniques

to solve both linear and nonlinear optimization problems. By adding barrier terms to the objective function, these methods ensure the search remains within the feasible region. To achieve the optimal solution, IPMs design a central path through this region. IPMs have many benefits, including performing the same no matter how big the network is or how many control variables are used. For example, [82] looks into how IPMs can be used to solve OPF problems using three algorithms: Multiple Centrality Corrections (MCC), Pure Primal Dual (PD), and Predictor Corrector (PC). The study evaluates the performance of these algorithms on various OPF problems, including minimizing generation costs, active power losses, maximizing load ability, and minimizing load shedding across different test systems.

The results show that higher-order IPMs, like the PC and MCC algorithms, do better than the algorithm in terms of the number of iterations and the amount of time needed to process them. The MCC algorithm also does a better job of convergent and robustness, which makes it a strong alternative to the PC algorithm. Study [83] introduced innovative versions of IPMs designed for tackling OPF problems. Using affine scaling and centralization together in a smart way is what the method does. It shows that this

combination makes the optimization process much faster and more reliable. The study also employs a perturbation parameter in the Karush-Kuhn-Tucker (KKT) conditions to address optimization problems in Nonlinear Programming (NLP). Recently, condensed-space interior-point methods in [84] have been used to quickly solve large-scale nonlinear programming problems, especially ACOPF problems that use GPUs.

C. GRADIENT METHOD

In the late 1960s, the the gradient method was one of the earliest techniques employed to address OPF problems. By leveraging the first-order derivative vector of the cost function (the gradient), these methods iteratively find directions that enhance the solution. Gradient methods ensure convergence for well-behaved functions. However, these methods tend to be slower compared to higher-order optimization methods.

In study [85], the traditional penalty function method was changed into the method of multipliers to create minimizing functions with constraints. This method effectively addresses numerical problems such as round-off errors, ensuring improved convergence for constrained optimization problems. Moreover, various gradient techniques, like the basic gradient method, Newton's method, and the conjugate gradient method, have been reviewed, highlighting their applications and computational challenges. While gradient methods are known for their simplicity and reliability, Newton's method provides faster convergence, albeit at the cost of increased computational demands. It has been shown that the multipliers method can get around the problems with regular penalty functions, which are a good way to solve optimization problems with limited resources.

The work in [86], a novel online gradient approach is described for solving OPF problems in radial networks. The program consistently meets power flow equations and operational limitations by changing control variables in real time. The approach allows for real-time optimization by implicitly solving power flow equations using the network. It is demonstrated to converge to local optima and, under certain conditions, global optima. Using linearized power flow equations cuts down on the amount of work that needs to be done, and the results are much faster convergence with little change in the final goal values. Extensive numerical testing on networks with up to 1990 buses demonstrates the algorithm's efficiency and applicability for real-time control of dispersed energy supplies.

D. QUADRATIC PROGRAMMING METHOD

Optimization technique characterized by an objective function formulated as a non-linear quadratic equation while all associated constraints are linear. In [87], a full quadratic model for OPF is shown. It has hierarchical structures, target functions, and constraints that can be changed to allow for personalized system control. The method uses sensitivity

analysis and optimal constraint adjustments to provide global optimum solutions. It includes special examples that reduce computational effort and allow for the selection of binding or nonbinding constraint sets. The flexible OPF package has been successfully tested on actual power systems, demonstrating its potential for real-time applications with utility companies.

In [88], a new QP approach to AC OPF designed for intelligent distribution grids is introduced. A new QP method was created to make the framework more durable and better at computing. It was tested against standard non-linear, quadratically constrained, and linearized formulations. The study focuses on two applications, which are optimizing the management of distributed generation units to maximize renewable energy usage and minimizing power losses by controlling capacitors. The QP method outperforms traditional methods, providing faster computational times and high-quality solutions.

A new method using quadratic programming to control dynamic direct current OPF that works well at different load levels and stays within the limits of the generator ramp rate [72]. This method incorporates previously overlooked constraints, such as transmission line capacity, into the economic dispatch framework. The effectiveness of this approach is validated through simulations using both a standard IEEE 14-bus test system and the complex Jawa Bali 500 kV 25-bus system. It shows that integrating DC load flow with quadratic programming provides robust solutions for managing real-time changes in load demand and renewable energy. These systems respond to operational limits such as ramp rates and transmission capabilities.

E. NEWTON-RAPHSON

The Newton-Raphson (NR) approach solves nonlinear problems by iteratively linearizing them. It starts with an initial estimation and then utilizes the function's differential to generate a linear approximation, improving the result with each iteration. This approximation converts the non-linear issue into a linear one at each step, allowing subsequent approximations to rapidly converge on the true root. The NR is highly appreciated for its quadratic convergence toward the solution, which implies that the error drops dramatically with each iteration. It depends on a good initial guess.

The NR method is extensively used for solving non-linear power flow equations ensuring that system voltages are maintained close to nominal values and that generator outputs and transformer tap ratios are efficiently calculated. The study, [89], describes a Newton-Raphson-based algorithm with sensitivity factors that improves real-time emission dispatch in power systems. This approach incorporates the Jacobian matrix and B-coefficients from the GGSD factor, increasing computational efficiency. It considerably decreases execution times while maintaining numerical accuracy, as evidenced by IEEE and TPC systems. The technique

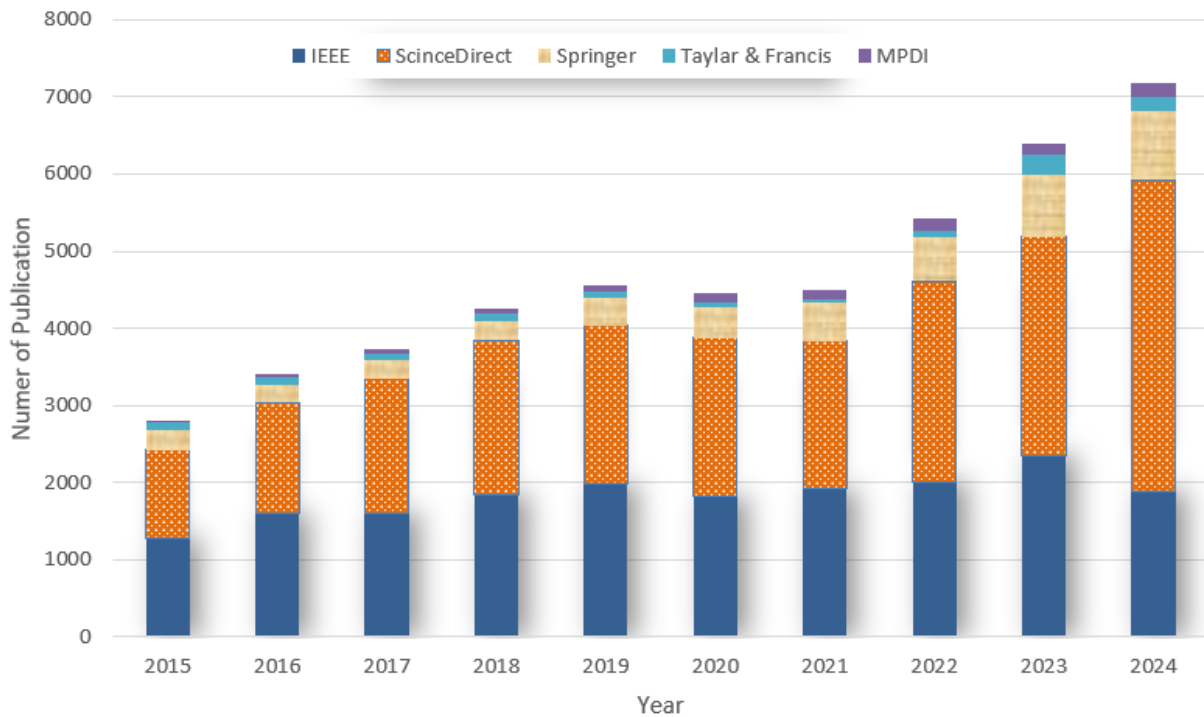


FIGURE 4. Recent publications on OPF across several databases.

effectively achieves both economic and environmental goals. It is well suited to real-time applications in power system operations.

In [90], a novel technique is introduced for solving the OPF problem by converting the KKT system into a collection of semismooth equations. This method controls inequality and limited constraints, resulting in fewer dual variables in the KKT system. The semismooth Newton algorithm exhibits excellent global and local convergence. Numerical tests on standard IEEE systems, such as the 9, 30, 57, and 118 buses, show that these algorithms are fast and good at solving OPF problems.

The traditional NR method for PF analysis can be improved by replacing the power mismatch equations with nonlinear current mismatch equations. This makes the formulation of the Jacobian matrix a lot easier [91]. The simplified NR method keeps the strong quadratic convergence properties of the standard NR method but makes computations faster by making derivative calculations less complicated. Simulations on different IEEE test systems, such as 5, 6, 24, 30, and 57-bus configurations, show that the simplified method converges just as well as the standard NR method, but it needs fewer iterations and less time to calculate.

IV. AI-BASED METHODS

Despite advancements in the development of classical optimization methods, they still face numerous shortcomings that hinder their optimal functioning in complex real-world power systems.

- 1) **Local Optimum Trap:** The methods will be stuck in local optima in solving of non convex problems.
- 2) **Single Solution Capability:** The method produce only a single global optimum solution per simulation of the runs.
- 3) **Linearization Requirement:** Some classical methods involve linearization of problems to make them easier to solve, but this in a way loses important aspect of highly nonlinear power system dynamics, thus generating less accurate solutions.
- 4) **Differentiability Requirement:** The OPF approaches utilizing conventional optimization techniques assume that the objective functions and constraints are differentiable.
- 5) **Differentiability Requirement:** Classical OPF techniques utilize first- and second-order derivatives of objective functions and constraints, necessitating that these functions and constraints be smooth or differentiable.
- 6) **Slow Convergence:** Convergence rates are slow, particularly for complex and large-scale systems, indicating that traditional methods are ineffective for rapid decision-making processes.
- 7) **Sensitivity to Initial Conditions:** Many conventional optimization techniques exhibit significant sensitivity to the first solution estimate and need substantial time to converge if the estimated solution deviates markedly from the actual solution.
- 8) **Inflexibility in Handling Multi-Objective Problems:** Conventional approaches are often designed for

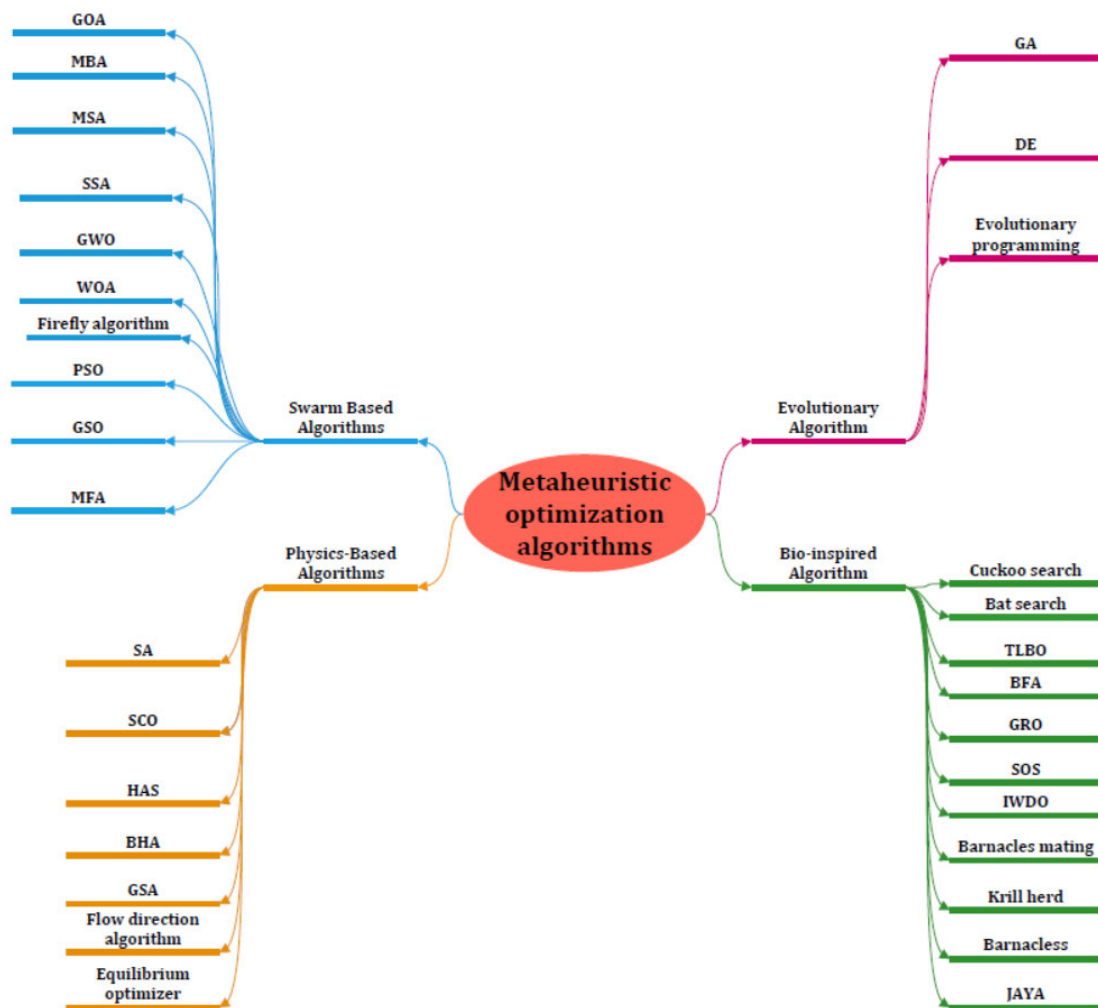


FIGURE 5. Various metaheuristic optimization algorithms for OPF.

single-objective issues and generally require additional adjustments or complex formulations to address multi-objective situations.

- 9) Limited Ability to Handle Uncertainty: Classical methods usually categorize inputs as deterministic and fixed which are not viable for power systems with uncertainty such as varying renewable resources or unpredictable loads.

These issues highlight the deficiencies of conventional OPF approaches and advocate for the adoption of advanced techniques, like metaheuristics and ML. Contemporary methodologies are more adept at addressing challenges like as nonlinearity, uncertainty, and larger systems efficiently.

A. SEARCH-BASED METHODS

The application of search-based optimization techniques has turned out to be significant in the past few years and has been widely used to solve OPF issues in complicated power systems, as shown in Figure 5. These techniques, derived

from natural and social processes, aim to effectively search large and complex solution spaces. It is also important to note that search-based methodologies are flexible and can easily solve non-linear, non-convex, and multi-objective optimization problems, making them ideal for power system optimization.

1) GENETIC ALGORITHM

The GA is a widely utilized and powerful approach for addressing a variety of optimization problems. GAs are techniques for optimization based on natural selection and genetic evolution. GAs are designed to explore difficult and large search spaces by mimicking biological processes like selection, crossover, and mutation in order to solve optimization problems. The possible solution encoded as a chromosome is evaluated by fitness functions, and the most successful solutions are passed along to the next generation [92]. That allows GAs to solve both constrained and unconstrained optimization problems. An improved GA (IGA) was created to help with the economic dispatch

problem for units that have valve-point effects and more than one fuel option [93].

The IGA incorporates an improved evolutionary direction operator and a migration operation which boost the algorithm's efficiency and solution exploration capabilities. To handle the complex constraints of economic dispatch a multiplier updating (MU) method is integrated ensuring that both equality and inequality constraints are met. This hybrid approach outperforms traditional methods providing robust and effective solutions for complex economic dispatch problems demonstrating showing its usefulness and efficiency in real-world power systems.

A novel method to improve the effectiveness of GAs for addressing the OPF challenges was introduced in [94]. This method focuses on picking the right initial complex voltages at the generator buses to keep constraint violations to a minimum. This cuts down on computation time and improves the quality of the solution. The proposed initialization procedure ensures efficient optimization by starting with control variables that are already close to optimal. Tested on standard IEEE test systems, the method shows competitive results compared to existing techniques. The GA-OPF is used in [95] to optimize the placement and operation of distributed generation (DG) units and energy storage systems in power distribution networks. The GA-OPF identifies the optimal locations, capacities, and operational strategies for both renewable and dispatchable DG units, as well as energy storage systems.

2) PARTICLE SWARM OPTIMIZATION

PSO was originally developed in 1995 [96]. It is a stochastic, population-based optimization technique that draws inspiration from the collective behavior of birds flocking or fish schooling. PSO uses a swarm of particles, each representing a potential solution, to explore the solution space. By sharing individual and collective experiences, these particles converge toward the optimal regions in the search domain. Each particle then finds and locates its new position based on the individual's and neighbors' previous experiences and the overall best experiences with special emphasis on the global best position. In order to obtain even better solutions, the algorithm changes the velocity and position of the particles employed in this case until the algorithm converges.

PSO was used in the power system [97] to tackle the ED problem. The study concentrated on units that exhibit valve-point effects, resulting in non-smooth input-output relationships. Simulations validate the performance of PSO on an IEEE 30-bus system with six generating units. Findings show that PSO delivers somewhat better results than Evolutionary Programming (EP), demonstrating its capability to manage complex, non-linear, and non-smooth optimization problems. Reference [98] present a new method to fine-tune control parameters in power systems to solve OPF; the approach combines fuzzy logic, GA, and PSO. The

goal is to lower the cost of active power generation while meeting various linear and nonlinear constraints. The hybrid method was tested on a modified IEEE 30-bus system with a particular focus on reducing fuel expenses. Results indicate that this integrated approach is more efficient and reliable than existing methods.

PSO was used for solving the OPF problem integrated with FACTS devices [99]. The method addresses various practical challenges such as valve-point effects, multiple fuel options, and restricted operating zones. The comprehensive-learning PSO (FBI-CLPSO) algorithm enhances its efficiency by employing a fuzzy-based approach. It was tested on the IEEE 30-bus system, which shows that it is more efficient and effective than traditional methods at solving the non-smooth and non-convex OPF problem. In [100], the PSO method successfully reduces the overall fuel cost for power generation to \$786.03 per hour, representing a 4.538% decrease compared to the robust Monarch Butterfly Optimization (MBO) algorithm. Modern optimization techniques differ from classical ones due to several key characteristics: they are applicable to both small and large-scale systems, they demonstrate high reliability in reaching optimal solutions, and they achieve convergence to the optimal solution more quickly than various other algorithms.

3) ANT COLONY OPTIMIZATION

The ACO was first proposed in 1996 [101]. This algorithm, inspired by the foraging behavior of ants, uses artificial agents (ants) that deposit pheromones on paths they travel, with the pheromone intensity influencing the probability of path selection by other ants. The method keeps getting better by telling the ants to come up with solutions using pheromone trails and their own knowledge of what works. This changes the pheromones to encourage successful pathways.

The study [102] presented the ACS method for addressing the OPF problem. The ACS method was used to find the best settings for control variables like VAR compensation blocks and transformer taps, as well as to get the best voltage levels while staying within the limits of the power system. The essential advantage of the method is its short execution time. When tested on the IEEE 30-bus system, the ACS algorithm demonstrated competitive performance and quicker execution times than alternative techniques. The Evolving Ant Direction Differential Evolution (EADDE) technique was introduced in [103]. This novel hybrid technique combines ACO to find the appropriate mutation operator for differential evolution (DE). While GA was employed to determine the ant colony's parameters. The EADDE algorithm was applied to IEEE 30-bus and 57-bus systems, aiming to reduce fuel costs, emissions, and power losses. The results showed that EADDE worked better than regular DE and other methods in terms of objective function values and convergence speed. It worked especially well with cost functions that were not smooth or convex. The Enhanced Ant Colony System (EACS) was developed to address

single- and multi-objective OPF problems, optimizing three different objective functions [104].

4) ARTIFICIAL BEE COLONY

The ABC algorithm was introduced in 2005 [105] based on how honey bees search for food. ABC employs three types of bees with different functions, including employable bees, onlooker bees, and scout bees. Employed bees find and gather information from known food sources and then share it with onlooker bees. The onlooker bees choose food sources to focus on based on this shared information. When existing food sources become depleted, scout bees randomly search for new ones. The ABC was evaluated using 14- and 30-bus IEEE systems [106]. The study showed that the ABC algorithm effectively minimized generation costs while adhering to power system constraints. The findings indicate that ABC outperformed GA and PSO in terms of both cost and computational time.

A modified ABC (MABC) approach is applied to tackle OPF using a fuzzy multi-objective optimization technique [107]. This algorithm aimed to reduce various objectives including fuel cost, pollution, power losses and voltage variations while effectively handling both discrete and continuous variables. The MABC method was tested on both IEEE 30-bus and 118-bus systems. The results show that the MABC algorithm was better than traditional approaches and other heuristic algorithms, like GA and PSO, because it got lower objective values and converged faster. The ABC method was used to solve the Optimal Reactive Power Dispatch (ORPD) problem in [108]. The goal was to lower actual power losses, lower voltage changes, and make voltage more stable. The ABC optimized different control parameters. The results based on tests on IEEE 57 and 30-bus systems indicated that the ABC algorithm provided both faster and better results than other techniques. In [109], the Non-dominated Sorting Hybrid Fruit Fly-Based Artificial Bee Colony (NSHFABC) algorithm was used to solve the multi-objective OPF problem in different IEEE systems. The goal was to lower generation costs, power losses, and severity indices while taking into account practical issues such as ramp rate limits. The results demonstrated that it outperformed other methods in terms of solution quality and convergence speed.

5) DIFFERENTIAL EVOLUTION ALGORITHM

The DE works through three basic mechanisms, which are mutation, crossover, and selection [110]. Mutation produces new candidate solutions by combining the weighted differences of two randomly generated vectors into a third vector. Crossover mixes mutant vectors with existing ones to keep the population diverse. Selection keeps vectors with higher objective function values, guaranteeing population improvement over time. The DE algorithm is used to solve the OPF problem in [111], which tries to find the best control variables to improve voltage profiles, fuel costs,

and voltage stability. The speed and solution quality of the DE algorithm were compared with traditional global optimization algorithms and heuristic algorithms such as PSO and GA to test the proposed approach on the IEEE 30-bus system. However, the study also acknowledges that the best results of DE can be sensitive to the control parameters under consideration and could call for calibration based on the tackled problem. Furthermore, scaling up and complicating power systems require some refinement or enhancement of the basic DE method.

Many traditional optimization methods work best with smooth, convex cost functions, so a modified DE is used to solve the OPF problem when irregular and non-convex generator fuel cost functions are used in the power system. Changes to the mutation rules are also made to the proposed MDE algorithm. These changes help the convergence process and make the solutions more accurate. The algorithm was tested on a six-bus system as well as on the IEEE 30-bus system with a variety of generator cost functions. The results suggest that the MDE algorithm delivers significantly better performance compared to the standard DE algorithm and other modern optimization methods in terms of solution accuracy and convergence speed. The EJADE-SP algorithm is a new improvement on adaptive differential evolution that is used to make power systems work better. The algorithm introduces four key enhancements: crossover rate sorting so as to collect more quality genes; parametric re-randomization to enhance search discontinuity; dynamic reduction of population size to increase convergence velocity; and re-balancing of penalty factors, adjusting them on their own. These improvements result in the efficient enhancement of the algorithm in determining an optimal solution. Compared to the proposed PSO, TLBO, and a variety of DE approaches when applied to a modified IEEE 30-bus system with conventional and renewable power sources, EJADE-SP came out on top. The IDE algorithm was created to solve the OPF multi-objective function [112]. It has three main changes: a random selection strategy for crossover, improvements to the trial operation process, and the addition of mutation calculations [113].

6) GREY WOLF OPTIMIZATION ALGORITHM

The GWO was introduced for the first time by Seyedali in 2014 [114]. It is an optimization algorithm derived from grey wolves lifestyles and their way of hunting. This optimization technique uses a model that has alpha, beta, delta, or omega wolves in replicating the observed leadership behavior and pack hunting strategies. The GWO algorithm mimics the way of surrounding the prey by adjusting the position of the wolves relative to the best solutions. Through adaptively changing the penalties or abortions, the algorithm sets a trade-off between how much the solution space is explored and how much is used in promising areas. With successive iterations, the position of the wolves becomes closer to the position of the prey. The tests show that the GWO works better than other algorithms like PSO, GSA, DE, FEP, and

CMA-ES in terms of how fast it converges and how well it solves problems.

For solving OPF, GWO and DE were employed in the study [115] to solve the problem for single and multi-objective functions. It has system constraints and uses a fuzzy-based Pareto front technique to find out the best compromise solutions for the multi-objective problems. The results from tests on both 30-bus and 118-bus systems show that both GWO and DE work well, with GWO giving about the same level of high-quality performance as DE. The proposed enhancement of GWO is called the Developed Grey Wolf Optimizer (DGWO), which is aimed at solving the OPF problem more effectively [116]. While using GWO, it has been found to be highly efficient, though the search is likely to be trapped in local optima since many of the wolves have less diversification. By adjusting the population diversity factor through random mutations during the exploration phases, DGWO improves search ability to address this problem. Secondly, it improves the exploitation phase by modifying the positions of the wolves in a spiral formation to identify the optimal solution. The operator used is the adaptive one that can successfully manage the exploration and exploitation processes during the iterations.

A new DGWO algorithm was suggested. It used goal functions like quadratic fuel cost, piecewise quadratic cost, and quadratic fuel cost with valve point effect. We compare the solution efficacy of the proposed algorithm with the IEEE 30-bus system, and the DGWO results outperform other meta-heuristic methods. The study [117] proposed a new high-performance Crisscross Search-based GWO, which improves the basic GWO by interrupting the searching process with a greedy approach and integrating the crossover operators of horizontal and vertical types into the leading wolves in constructing the rich population. These modifications effectively speed up convergence and prevent stalling at local optima. The IEEE 30- and 118-bus systems are used to test CS-GWO. It works better and costs less than the basic GWO and a number of other metaheuristic algorithms.

7) OTHERS METHODS

Rao developed the Jaya Algorithm in the year 2016 [118]. The Jaya algorithm is one of the most effective optimization methods due to its simplicity and lack of parameters. The principle behind Jaya is straightforward. It continues to modify candidate solutions until it improves, retrieves the best one, and, at the same time, avoids the worst one in the population. In contrast to many other optimization algorithms, the Jaya Algorithm does not require any parameters to be set besides the population size and the maximal number of generations. Many works involving OPF have implemented the Jaya algorithm to optimize various objective functions. [119] used the Jaya algorithm to lower the cost of generation and active power loss, as well as to improve voltage stability by adding Distributed Generation (DG) resources and a sensitivity index to IEEE 30 and 118 bus systems. The authors

of [120] suggested the modified Jaya algorithm (MJAYA) as a way to solve OPF problems. The goal was to lower fuel costs, emissions, and transmission power losses while also improving voltage profiles, especially when renewable energy sources were added.

The Flower Pollination Algorithm (FPA) is a nature-inspired optimization method created by Xin-She Yang in 2012 [121]. It takes cues from the way flowers pollinate, simulating both biotic pollination (with the help of insects and animals) and abiotic pollination (via wind and water). By using these natural processes, the algorithm effectively searches for the best solutions, balancing between wide-ranging exploration and detailed focused searching. Modified FPA to address the OPF problem aimed to minimize fuel costs while satisfying power demand and system constraints. The MFPA enhances the standard FPA by optimizing the initial population and adjusting the local and global pollination switching process [122].

The Clonal Selection Algorithm (CSA) is an optimization process first described by Leonardo N. De Castro in 2002 [123]. CSA reflects on how the immune system produces antibodies by cloning and mutating depending on the antigens present in the system. Reference [124] designs a Multi-Objective Adaptive Clonal Selection Algorithm (MOACSA) for minimizing generation costs, transmission losses, and voltage stability issues in systems possessing FACTS devices and non-certain loads.

Kaveh introduced Charged System Search (CSS) in 2010, and it is an optimization approach that is based on electrostatic and Newtonian physics concepts. Each particle represents a possible solution that interacts with others using Coulomb's law [125]. The Reserve Constrained Dynamic Optimal Power Flow (RCDOPF) problem is about finding the best way to minimize operational costs while keeping a number of constraints in mind. The Enhanced Charged System Search Algorithm (ECSSA) is a good algorithm that has been used to solve this problem. It is clear from these results that the ECSSA works better than the others in both the small-scale power system (IEEE 30 bus system) and the large-scale power system (IEEE 118 bus system) [126].

A metaheuristic optimization algorithm called Tabu Search was created in 1980. It can be used to solve both combinatorial and continuous optimization problems [127]. This method uses adaptive memory structures to explore the search space while avoiding cycles. By maintaining a tabu list, it keeps track of recent changes and prevents them from being reverted, promoting the discovery of new options. The method has been used extensively to solve the OPF problem in [128] and [129]. The TS algorithm's flexible memory structure helps prevent local optima entrapment and enhances solution quality.

Simulated Annealing (SA) is an algorithm based on the process of annealing commonly used in metallurgy. SA was originally formulated by Kirkpatrick in 1983 [130] to help find an approximate global optimum for a specific function. The algorithm mimics the cooling process in metals,

where a material is heated and then gradually cooled to eliminate defects and achieve a state of minimum energy. SA begins with an initial solution and explores the solution space through random perturbations. These perturbations may sometimes result in worse solutions, which are accepted based on a probability that decreases over time, allowing the algorithm to avoid getting stuck in local minima. Several studies have shown that Simulated Annealing works to solve the OPF problem in a number of different IEEE bus systems [131], [132].

The Invasive Weed Optimization (IWO) algorithm was introduced in 2006 and mimics the natural behavior of weeds to address optimization problems [133]. The IWO starts with a random distribution of seeds that reproduce based on their fitness. Offspring are dispersed in the search space with a spread that narrows over time to ensure convergence to optimal solutions. Reference [84] proposed the Chaotic Invasive Weed Optimization (CIWO) algorithm incorporating chaos theory and assesses its performance in fine tuning the control variables using the OPF problem with non-smooth and non-convex fuel cost function of the generators. The effectiveness of the CIWO algorithm is analyzed for solving the IEEE 30-bus test system with reference to different objective functions.

The BBO is an evolutionary algorithm that was introduced by Simon in 2008 and inspired by the study of species distribution [134]. The algorithm views solutions as habitats, utilizing immigration and emigration rates to generate new candidate solutions. Unlike gradient-based methods, BBO does not require differentiability and is applicable to various optimization problems. Solutions are treated as habitats with varying quality, termed Habitat Suitability Index (HSI). High HSI solutions share their features with low HSI ones through migration while mutation introduces diversity. A number of studies [135] applied the BBO algorithm to OPF problems with both convex and nonconvex fuel costs. Simulation Conducted on the IEEE 30-bus test system, it seeks to reduce fuel costs and optimize voltage levels.

Civicioglu designed the Backtracking Search Algorithm (BSA) in 2013 [136]. It is an evolutionary optimization approach that uses differential evolution and backtracking to explore and exploit the search space. To maintain diversity and prevent getting stuck in local minima, BSA uses processes such as initialization, mutation, crossover, selection, and periodic backtracking. The BSA successfully handled the OPF in four different cases [137].

The main reason for comparing the different heuristic strategies created to solve OPF problems on IEEE systems 30-bus, 57-bus, 118-bus, and 300-bus in Tables 1, 2, 3, and 4 is to see how well they work based on the set of selection criteria, which include minimizing costs, reducing computational time, and being able to handle larger systems. Compared in this manner, the advantages and drawbacks of

different types of methods are discovered where they are all compared under the same standard, hence giving an insight into what sort of method is best suited for systems of a certain size and complexity.

B. DATA-BASED

Among data-based methods, machine learning is a particularly important type. In the previous decades, the industry did not fully embrace the use of ML due to limited availability of data and limited computational capability. Improvements in computing technology and the installation of advanced metering infrastructures that help in data collection are some of the enabling factors that have led to many successful uses of real-world applications [173]. The recent growth in ML techniques combined with better computing power and more available data has led the power systems community to use ML to solve some complex problems, such as OPF [174], [175], and ED problem [176], [177]. ML is a powerful tool for analyzing data and giving system operators important insights to make quick and effective decisions. Study [178] explores four primary machine learning approaches to address ACOPF problems. ACOPF problems can be solved directly with advanced deep learning methods, models can be trained offline with historical data to get close to optimal solutions without direct optimization, machine learning can be used to predict initial conditions that make traditional solvers more efficient, and physical constraints can be added to neural network training to make it more robust and make sure that solutions respect system limits. Figure 6 illustrates the number of papers published on different types of ML methods used to solve OPF between 2015 and 2024 in IEEE Xplore. The results show that supervised learning has the highest number of publications.

1) SUPERVISED LEARNING

In supervised learning, a model is trained using labeled datasets where each input is associated with its corresponding output, allowing the model to learn the correct relationships between inputs and outputs. The training enables the model to predict an output from new inputs in tasks such as classification and regression [177], [179], [180]. Different types of supervised ML algorithms are used to solve OPF problems.

The study [181] suggested a data-driven linear robust OPF model to deal with the problems caused by the fact that distribution networks (DNs) are complicated and change over time, which makes it hard to keep accurate physical models. The method leverages historical data and linear regression to predict power flows and voltage levels while incorporating model errors as uncertainty variables using kernel density estimation. Reference [182] presented a new way to solve the complex OPF problem through a machine learning technique, which is Stacked Extreme Learning Machine (SELM). This SELM simplifies the problem by

TABLE 1. Minimization of total fuel cost for IEEE 30-bus.

Number of Reference	Algorithm Name	Total Cost (\$/h)	Iterations	Time (s)	Type of PC
[100]	Developed Particle Swarm Optimization (PSO)	786.03	30	–	–
[139]	AMTPG-Jaya Algorithm	800.1946	34	14.603	2.7 GHz processor, 16 GB RAM
[140]	Improved Bird Swarm Algorithm (IBSA)	800.3975	–	–	Intel Core i7, 8 GB RAM
[141]	Semidefinite Programming (SDP)	801.8888	50	–	Intel Core i7, 2.90-GHz, 12 GB RAM, 64-bit
[142]	Fitness-Distance Balance Based Teaching-Learning-Based Artificial Bee Colony (FDB-TLABC)	785.5850	–	–	Intel Core i7, 8 GB RAM
[143]	Space Exploration and Unimodal Region Elimination (SEUMRE)	798.8878	–	–	Intel Core i7, 16 GB RAM, 2.7 GHz
[144]	Improved Archimedes Optimization Algorithm (IAOA)	799.068	400	0.22	Intel Core i7-1065, 2.8 GHz, 8 GB RAM, 64-bit
[145]	Multi-Objective Search Group Algorithm (MOSGA)	799.1777	–	–	–
[146]	Efficient Parallel Genetic Algorithm (EPGA)	800.9617	–	–	2.6 GHz Pentium-IV PC
[147]	Flow Direction Algorithm (FDA)	802.136	200	–	Intel CORE i5, 9th Gen, 8GB RAM
[148]	Enhanced Genetic Algorithm (EGA)	799.56	50	–	–
[149]	Efficient Evolutionary Algorithm (EEA)	800.0831	109	5.9694	PC-Core2 Quad computer, 3.24 GB RAM
[150]	Tabu Search Algorithm	\$800.41	45	20	Intel Pentium III, 500 MHz, 128 MB RAM
[151]	FPSO	\$800.72	300	–	–
[152]	Evolving Ant Direction PSO	\$800.2276	–	–	Pentium-IV, 2.66 GHz, 512 MB RAM
[153]	Artificial Bee Colony (ABC) Algorithm	\$800.66	50	10.2	3 GHz Intel Processor, 2 GB RAM
[154]	Modified Shuffle Frog Leaping Algorithm (MSFLA)	\$802.287	–	–	Pentium IV, 2.66 GHz, 512 MB RAM
[155]	Hybrid MICA-TLA	\$801.0488	–	–	Pentium IV E5200, 2 GB RAM
[156]	Hybrid Cuckoo Search Algorithm	\$802.9293	100	–	i3-370M processor 2.40 GHz, 3 GB RAM
[157]	Multi-objective Harmony Search (MOHS)	\$798.8000	–	–	–
[158]	Ant Colony Search (ACS) Algorithm	\$804.86	100	5.90	2x Intel Xeon W3520 2.67GHz, 3GB RAM
[159]	Black-Hole-Based Optimization (BHBO)	\$799.9217	–	–	2.20 GHz i7, 8 GB RAM
[160]	Quasi-Oppositional TLBO	\$799.15	–	–	Pentium IV, 2.66 GHz, 512 MB RAM
[161]	Teaching-Learning-Based Optimization (TLBO)	\$799.0715	–	0.92	2.20 GHz i7, 8 GB RAM
[162]	Multiagent Based Differential Evolution (MADE)	\$798.73	–	–	–
[163]	Differential Evolution Algorithm (DEA)	\$799.2891	–	–	Pentium IV, 1.3 GHz
[164]	Modified Differential Evolution Algorithm (MDE)	\$802.376	160	23.25	Pentium IV, 3.0 GHz, 512 MB RAM
[165]	Whale Optimization Algorithm (WOA)	\$799.367	500	–	2.60 GHz i5, 4 GB RAM
[166]	MFPA	799.1592	300	–	Intel Core i7 4770, 3.4 GHz, 8GB RAM
[167]	Hybrid CIWO Algorithm	800.4735	–	–	Pentium IV E5200, 2GB RAM
[168]	PM	799.0566	100	10.71	Pentium IV, 2.3 GHz

breaking it into three stages: first, it learns the power flows in the network; second, it figures out the voltages at different

points; and third, it fine-tunes the control settings and overall costs. By pre-classifying data samples, the method

TABLE 2. Minimization of total fuel cost for the IEEE 57-Bus System.

Number of Reference	Algorithm Name	Total Fuel Cost (\$)	Iterations	Time (s)	
[152]	Evolving Ant Direction Particle Swarm Optimization (EADPSO)	41,697.54	150	12.23	Pentium-IV, 2.66 GHz computer with 512 MB RAM
[153]	Artificial Bee Colony (ABC) Algorithm	41,693.9589	200	226.23	Intel 3 GHz Processor, 2 GB RAM
[169]	Gravitational Search Algorithm (GSA)	41,695.8717	–	400	2.63 GHz Pentium IV, 512 MB RAM
[170]	Imperialist Competitive Algorithm (ICA) with Modifications	41,709.7292	–	–	Pentium IV, 2 GB RAM
[171]	Adaptive Group Search Optimization (AGSO)	38,218.856	300	96.32	Intel Core i5, 2.5 GHz, 4 GB RAM
[172]	Improved Bird Swarm Algorithm (IBSA)	41,663.5500	1000	–	Intel Core i7, 2.40 GHz, 8 GB RAM

TABLE 3. Minimization of total fuel cost for the IEEE 118-Bus System.

Number of Reference	Algorithm Name	Total Fuel Cost (\$)	Iterations	Time (s)	PC Type
[119]	Jaya Algorithm	129,490.54	300	1.83 per iteration	Intel Core i7, 16 GB RAM, 2.7 GHz
[161]	Teaching-Learning-Based Optimization (TLBO)	129,682.844	–	–	2.20 GHz Intel Core i7, 8 GB RAM
[173]	Biogeography-Based Optimization (BBO)	129,686.00	100	29.8 per iteration	–
[174]	Multi-Operator Differential Evolution (MODE)	134,993.01	100	–	16 GB RAM, a core i7 processor with a 3.4 GHz and Windows 10
[175]	Modified Log-Barrier primal-dual interior/exterior-point method	135,202.95	18	22.41	Intel Core i3 processor, with 4 GB of RAM, and Windows 10
[176]	Modified JAYA (MJAYA) Algorithm	128,483.82	450	2459	2.8-GHz i7 PC with 16 GB of RAM

TABLE 4. Minimization of total fuel cost for the IEEE 300-Bus System.

Number of Reference	Algorithm Name	Total Fuel Cost (\$)	Iterations	Time (s)	PC Type
[177]	GA	805,814.61	–	–	Core 2 Quad Computer with 8 GB of RAM
[178]	tri-level decomposition strategy	717,413.00	–	–	Intel(R) Core(TM) i7 2.50 GHz CPU, and 8 GB of memory
[179]	A gradient-based approach	724,921.22	42	283.06	Intel Core i7, 16 GB RAM, 3.5 GHz
[180]	EGA	834,442.67	200	823.0461	Core 2 Quad Computer with 3.24 GB of RAM
[181]	Combined Efficient Metaheuristic Approach	834,801.53	200	91.1791	Core 2 Quad Computer with 3.24 GB of RAM
[182]	Evolutionary Algorithm (EEA)	834,986.73	200	154.9357	–
[183]	GSO	859,834.82	–	–	–

makes the learning process more efficient and accurate. A supervised learning method named Compact Learning was introduced to improve the scalability of ML models in OPF problems [183]. This method employs Principal Component Analysis (PCA) to reduce the dimensionality of the solution space, allowing the model to learn more efficiently in a compressed subspace. By decreasing the number of parameters to train, this approach addresses the scalability issues typically encountered with large-scale power systems. Using the Generalized Hebbian Algorithm (GHA) to learn both the main parts and the path from inputs to solutions speeds up the learning process. The model has steps for post-processing that involve solving a PF problem

and using the predictions to warm-start traditional OPF solvers. This is done to make sure that the predictions are both physically and engineeringly possible. One of the most popular ML models for solving OPF problems is the NN. Each node connects to others in adjacent layers, forming a network of neurons. These connections have weights that adjust to the learning process based on training data. The process involves feeding input data through the input layer, and then it gets processed in each layer. Different types of NN have been used to solve OPF. Using a deep neural network (DNN) approach to solve AC-OPF problems, the DNN forecasts essential operating variables and then computes the remaining ones by solving power flow equations, thereby

Number of Publication in IEEE and ScienceDirect

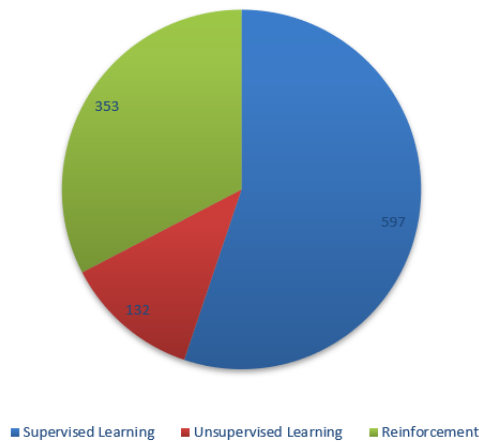


FIGURE 6. Number of publications on OPF ML methods in IEEE and Elsevier from 2015 to 2024.

ensuring power balance in the system. The training process incorporates a penalty method to manage constraints and ensure viable solutions. The model was trained using a dataset of 12,500 samples with the data divided into 80% for training and 20% for testing [184].

CNN was used to address the AC-OPF [185]. The method uses cluster analysis to figure out the topology of the network and how it works by looking at old data. This lets the CNN accurately guess voltage outputs in a range of situations. The model Trained on 8000 samples and evaluated on 2000 samples, the results demonstrate a 350x gain in speed over traditional methods. This study [186] employed a physics-guided graph convolution neural network (GCNN). The GCNN does a good job of capturing the complex nonlinear relationships and topological features of power systems because it embeds power flow equations directly into the graph convolution process. The method also incorporates a feature construction strategy that includes physical constraints and a specialized loss function to maintain accurate physical relationships between outputs. The model was trained on 10,000 samples and tested on 2,000 samples, all representing power systems as graphs. This training process shows that the GCNN improves convergence and prediction accuracy under varying topologies with an average accuracy increase of 13.30% and up to 32.63%.

A Recurrent Neural Network (RNN) is introduced as an innovative approach for addressing the OPF problem in distribution networks [187]. The methodology involves creating a dataset using a continuation power flow method, which is then employed to train the RNN to predict the relationship between load and generation output. The study utilizes a dataset for a 42-bus distribution network system. This dataset contains a total of 1002 samples, with 800 samples used to train the RNN model and 202 samples reserved for testing.

2) UNSUPERVISED LEARNING

Unsupervised learning is one of the subcategories of machine learning where the model is trained on unlabeled data. That means the model is only provided with input data and is expected to look for patterns and structures on its own. Traditional solvers can be very slow and resource-intensive when dealing with constrained optimization, especially for large-scale and urgent problems like ACOPF. The method “Deep Lagrange Dual with Equation Embedding (DeepLDE)” used in [188] falls under the category of unsupervised learning in machine learning. This approach does not rely on labeled data but instead uses constraints and optimization techniques to train the model, making it suitable for complex problems like AC-OPF. This method significantly increases computational speed up to 35 times faster than conventional approaches (non-convex).

In [189], a new method was introduced of unsupervised learning techniques to solve OPF, which is Generative Adversarial Networks (GANs). The GAN works with a generator creating potential solutions while a discriminator checks their feasibility. This approach, guided by mutual information principles and electrical domain knowledge, ensures the solutions are cost-effective and meet necessary constraints. By training on feasible datasets, the method quickly produces near-optimal solutions, significantly improving computational speed. The DeepOPF-NGT (No Ground Truth) is used to deal with AC-OPF. It does this by using DNNs to learn how different load configurations affect bus voltages, without needing optimal solutions to be known ahead of time [190]. It incorporates a specially designed loss function that merges the objective of the AC-OPF problem with penalties for constraint violations and unmet loads. The method employs Kron reduction to manage zero-injection buses and an adaptive learning rate algorithm to dynamically balance gradient contributions. DeepOPF-NGT shows excellent performance, achieving feasibility and speed improvements on par with the best supervised methods.

3) REINFORCEMENT LEARNING

RL is a field of ML in which an agent improves its decision-making ability by interacting with an environment and taking actions that seek to maximize the overall reward it accumulates. Unlike the supervised learning technique that uses the data containing input-output pairs to train the model, or the unsupervised learning that explores the data in an attempt to discover specific patterns, RL focuses the agent’s learning on trial-error within the environment for enhancing the actions under the feedback it gets. A new approach was presented for real-time OPF that incorporates qualitative guidelines from Grid Codes and Operation Manuals [191]. Deep Reinforcement Learning (DRL) is used in this method, and it is improved by a GPT agent. The GPT agent sees these natural language instructions as rewards during training. By adding the GPT agent, the system can make the best use of power flow while taking both numerical limits and policy

TABLE 5. Summary of Machine Learning Applications in OPF.

Type ML of Application in OPF	Use	Examples	Advantages	Challenges	Applications Summary
Prediction and Approximation	Predicting initial conditions, Approximating solutions, Predicting parameters.	DNN, CNN, RNN	Fast inference, Handling large data sets, Improves computational efficiency.	Need labeled data, Potential constraint violations, Overfitting risks.	Predicting initial conditions and parameters for traditional solvers. Approximating solutions to reduce complexity and computation time. Learning from historical data to provide quick solutions.
Pattern Recognition and Anomaly Detection	Handling uncertainties, Exploring latent structures, Clustering, Anomaly detection.	Clustering, GANs	No labeled data needed, Discovery of hidden patterns, Reduces dependency on labeled data.	Model interpretability, Optimization difficulties, Difficult to integrate with traditional OPF.	Clustering similar network states to identify patterns. Anomaly detection and handling uncertainties without labeled data.
Dynamic Control and Optimization	Dynamic optimization, Real-time decision making, Adaptive control, Learning optimal policies.	Q-Learning, Actor-Critic	Learning from environment, Adapting to changing conditions, Handles complex decision-making.	High computational cost, Need for well-defined rewards, Potential safety concerns.	Dynamic optimization and real-time decision-making for control actions. Adaptive control under varying conditions. Learning optimal control policies.
Constraint Handling and Feasibility	Incorporating physical constraints, Ensuring feasibility, Combining with optimization methods.	PINNs, Hybrid Models	Ensures physical validity, Improves robustness, Enhances robustness and scalability.	Complex model formulation, Training difficulty, Increased training and validation efforts.	Incorporating physical constraints into learning process. Ensuring feasibility by embedding power flow equations in neural networks.

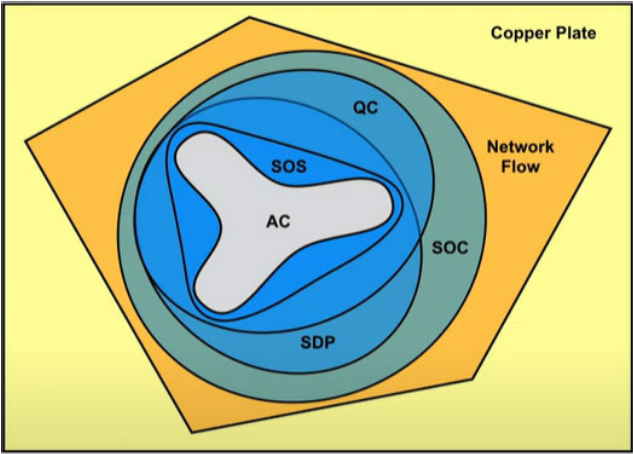


FIGURE 7. Convex relaxation of ACOPF.

requirements into account. Looking ahead, further research could explore the security aspects of these LLM-based data-driven methods and the development of domain-specific LLMs (large language models) for various power system tasks. It would be beneficial to consider and test a broader range of linguistic stipulations.

The work in [192] introduced a novel approach called hybrid knowledge-data-driven reinforcement learning (KDD-RL) to address real-time sequential OPF challenges. The method enables quick decision-making without requiring highly precise environmental parameters. By combining expert knowledge with a way to find constraints that can't be relied on, a better linear projection model is created to guarantee action safety. This improves system security during the initial stages of training and when unexpected events happen. This strategy enhances decision-making

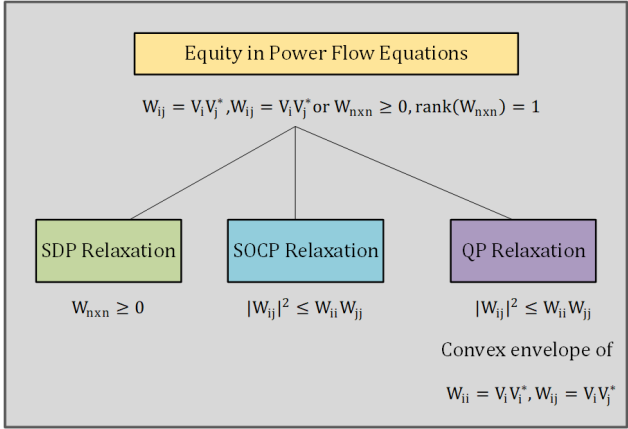


FIGURE 8. Power flow relaxation techniques: SDP, SOCP, and QP.

efficiency and ensures the secure operation of power systems.

ANNs are trained via reinforcement learning [193]. A high-performance batched power flow solver is used as the training environment for this method. It also looks at an improved loss function that has both objective terms and penalties for breaking constraints. This enables successful training without relying on reference OPF solutions and handles discrete choice factors such as transformer tap locations. The approach provides accurate solutions quickly and efficiently. Convex constrained soft actor-critic (CC-SAC) deep reinforcement learning (DRL) algorithm is also used to tackle real-time OPF challenges [194]. The method takes expert knowledge and a way to find inactive constraints and combines them to make a better model that makes sure safe actions are taken and system

security is improved during training and in unplanned situations. A new way to solve the look-ahead ACOPF problem was described in [195] (also refer FIGURE 7). It involves using model-informed reinforcement learning along with a physics-informed neural network based on the DDPG algorithm. The method fine-tunes active power outputs by calculating each generator's adjustable capacity and distributing deviations proportionally using the equal incremental rate principle.

TABLE 5. summarizes the applications of ML in OPF as discussed in the paper and different studies [33], [178].

V. OPF CONVEX RELAXATION METHODS

There are several benefits to using convex relaxation methods to solve OPF problems. These methods involve turning non-convex problems into convex ones so that any local optima that are found are also global optima. This gets rid of the risk of getting stuck in local optima. These methods are computationally efficient using standard convex optimization algorithms, which decrease the time and cost required for solving large-scale problems, enabling real-time applications. Convex relaxation methods make it easier to find workable solutions by giving you a good place to start. They also let you model a lot of different constraints and objective functions, such as generator constraints and voltage constraints, which are useful in real life. Convex relaxation, on the other hand, makes the initial problem's feasible area bigger. This has two big benefits: if the relaxed version is impossible, then the initial problem is also impossible. The best answer to the relaxed problem also gives a lower bound for the original problem, which lets us figure out how far away a possible solution is from the best result. This Figure 8 illustrates the relaxation methods applied to power flow equations [33].

A. SEMI-DEFINITE PROGRAMMING RELAXATION

The OPF problem was solved for the first time by using semidefinite programming (SDP) in [196]. This method represents a significant advancement in power systems optimization by transforming the traditionally nonconvex OPF problem into a convex problem, allowing the use of primal-dual interior point methods (IPM). The SDP approach eliminates the necessity for deriving Jacobian and Hessian matrices, which are required in conventional nonlinear programming (NLP) methods. The SDP method was tested by implementing it on power systems of varying sizes ranging from 4 to 300 buses. Two key conditions were identified that ensure the accuracy of SDP relaxation for solving OPF problems [197]. First, it was established that the SDP relaxation is exact if it produces a rank-one solution meaning the problem can be solved optimally. Second, it was determined that the duality gap is zero if the semidefinite matrix in the dual SDP solution has a zero eigenvalue with a multiplicity of 2. These two conditions are significant because they provide a clear way to verify when the SDP approach gives a precise and optimal solution.

In the study [198], the differences between the SDP relaxation solution and OPF model solution were also compared. Based on the results, it looked like the SDP relaxation usually gave the best bounds among the relaxations that had optimality gaps less than 1%. However, the optimality gaps varied widely across different test cases. In some scenarios, the gaps for the SDP relaxation reached up to 30%, particularly when numerical challenges arose or the solver did not converge. Overall, SDP relaxation works well but is proven to have non-guaranteed minimal optimality gaps in all situations. By incorporating convex envelopes, bound tightening, and valid inequalities, we significantly improved the accuracy of SDP relaxation for solving OPF problems. The result showed that the hybrid SDP relaxation could reduce the optimality gap to less than 1% in 63 out of 71 test cases [199]. The SDP was used to solve multi-objective OPF problems [200]. This model tackles multiple goals, such as reducing fuel costs, transmission losses, and emissions, by using the ϵ -constraint method to turn these goals into constraints and create a range of optimal solutions. Their study tested this method on several systems, including the IEEE 30, 57, and 118-bus systems, and an Indian utility 62-bus system.

The study [201], addressed the computational challenge and inaccuracy associated with conventional OPF models when applied to modern distribution networks with high integration of distributed energy resources (DERs). Traditional linear and non-linear OPF models either fail to provide feasible solutions or lack accuracy for large networks, and they do not guarantee globally optimal solutions due to the non-convex nature of the power flow equations. To solve this problem, the study proposes a novel approach using SDP to create OPF models that are both scalable and accurate for radial distribution networks. The SDP-OPF model utilizes a bus injection model (BIM) for single-phase networks and a branch flow model (BFM) for multi-phase unbalanced networks.

B. SECOND-ORDER CONE PROGRAMMING RELAXATION

SOCp serves as an efficient method for initially addressing radial distribution load flow problems [202]. SOCp addresses numerical ill-conditioning and ensures polynomial time complexity using interior-point methods. Nine distribution systems validated the method, demonstrating its efficiency and robustness in comparison to traditional methods. The results indicated that the conic optimization solutions were nearly identical to those obtained from Cespedes method but with improved numerical stability and computational efficiency. The relaxation has proven to be accurate in radial networks provided that there are no upper bounds on power injections. A branch flow model was presented [203], focused on improving both radial and mesh power systems via a two-step relaxation approach. First, voltage levels and phase angles were removed, and second, the problem was converted into a conic formulation. The relaxation has been shown

to be accurate in radial networks provided that there are no maximum constraints on power injections. The branch flow model allows us to formulate the OPF problem as a convex optimization problem, which interior-point methods can effectively solve.

An approach for enforcing cycle constraints was introduced, ensuring that the sum of voltage angles throughout a cycle is zero, addressing the issue of non-convexity in cycle-based networks [204]. Using second-order moment relaxation matrices this technique improves the accuracy of SOCP relaxation for radial networks while also reducing the computational burden. An algorithm is proposed for computing approximate solutions of the SOCP relaxation of OPF problems in mesh networks [205]. The OPF problem is transformed into a difference-of-convex programming (DCP) formulation and then solved by using the penalty CCP approach. This method begins with the solution of a tightened SOCP OPF as a starting point and seeks to find a global or near-global optimal solution. This method takes advantage of both SOCP and CCP in handling the non-convexity in the OPF problems without having to compromise computational speed. SOCP has trouble with OPF models in meshed power transmission networks; it doesn't always show voltage angle differences correctly because it doesn't have the right cyclic constraints [206]. This leads to inaccuracies and inefficiencies in power flow solutions. To address this issue, we proposed an enhanced SOCP-OPF model that incorporates a convex envelope to accurately represent bus voltage angles, ensuring that cyclic constraints are met for all mesh cycles in the network.

C. QUADRATIC CONVEX RELAXATION

QC relaxation can be considered an improvement over SOCP relaxation in terms of providing tighter bounds and more accurate solutions ([33]). The study [198] also found that QC relaxation is more accurate and stable than Second-Order Cone (SOC) relaxation, especially when there are small differences in phase angle. One of the study's significant findings is that QC relaxation is more computationally efficient than SDP relaxation, making it quicker. Extensive computer tests showed that QC relaxation is both practicable and beneficial for real-world power system optimization.

The study [207] suggested a new improved quadratic convex (QC) relaxation of the AC-OPF problem and an optimization-based bound-tightening (OBBT) algorithm. This algorithm is designed to calculate tight and feasible bounds on voltage magnitudes at every bus as well as the phase angle difference between the two buses connected by the branches in the network. Analytic assessment corroborates the actual advantages of this strengthened QC relaxation relative to other variants of QC. A lot of numerical testing of the suggested method is done on standard AC-OPF networks, and it always performs better than the other methods tested by having the smallest variable bounds and optimality gaps while taking less time to run. An enhanced QC relaxation

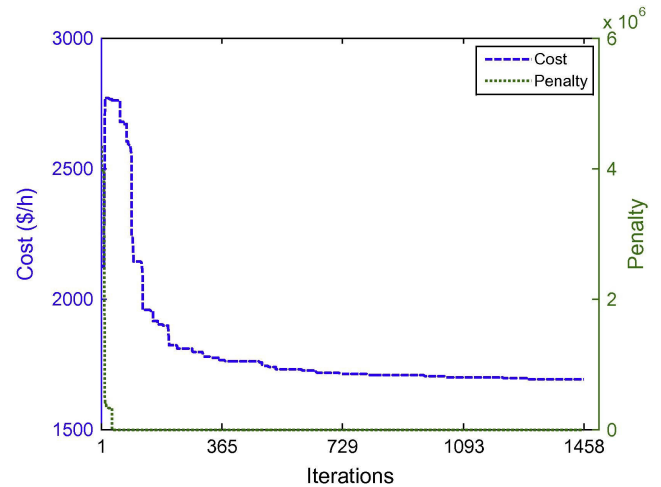


FIGURE 9. Fuel cost variation for Case-1: Optimization of Cost Function [209].

was introduced for AC-OPF problems, featuring two key improvements [208].

For the first, branch admittances are shown in both polar and rectangular ways. For the second, trigonometric envelopes are made tighter by using a coordinate transformation with complex per unit base power normalization. Empirical analysis across various test cases validates these improvements, demonstrating superior performance compared to existing QC formulations with minimal impact on computational efficiency.

VI. CASE STUDY FOR OPF

This explores case study from [72] and [209] to show how well the League Championship Algorithm and Whale Optimization work. These techniques have been tested on two different test systems: the Algerian test system, which has 59 buses, and the New England test system, which has 39 buses, 46 branches, and 10 generators. The tests also had non-convex constraints like generating capacity, ramp limits, voltage limits, and no-operating zones. The Algerian 59-bus test network consists of 59 buses, 10 generators, 36 loads totaling 684.10 MW, and 83 branches. This study examines two scenarios of OPF. The first objective is the minimization of generating fuel costs, while the second objective is the minimization of non-convex costs and the reduction of emissions, as presented in equations (2) and (6).

A. CASE-1: OPTIMIZATION OF COST FUNCTION

This scenario will evaluate fuel cost as the goal function to reduce energy production expenses while adhering to generating constraints, including generation limitations and ramp-rate restrictions of generators. The cost coefficients of the objective function a , b , and c for this system are detailed in the Appendix of [209].

FIGURE 9 illustrates the progression of the objective function across the iterations. It is important to note that in

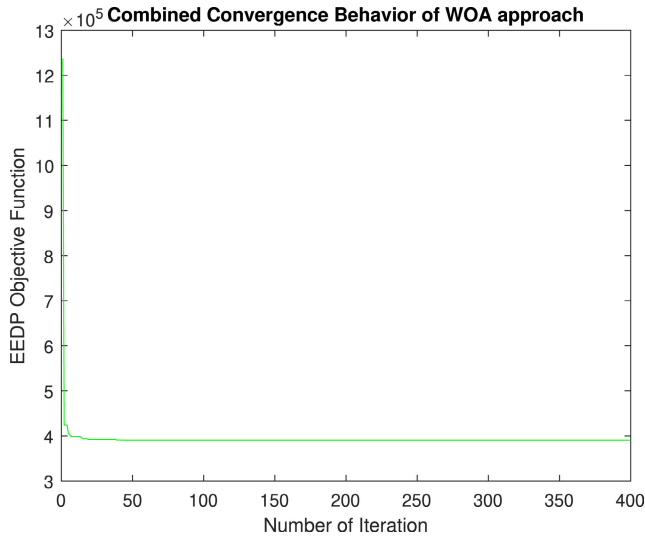


FIGURE 10. Convergence for Case-2: Optimization of Combined Cost and Emission Function [72].

this context, the fuel cost and the penalty term, along with their summation, form the objective function. The data shown in FIGURE 9 demonstrates that the suggested League Championship Algorithm has exceptional convergence capability and excels in achieving optimal cost. The suggested strategy rapidly converges to the global optimum, demonstrating its effectiveness in identifying the global optimal solution in less iterations, while also requiring less processing time.

B. CASE-2: OPTIMIZATION OF COST AND EMISSION FUNCTION

This case will consider the fuel cost a multi-objective function to minimize the energy production cost as well as emission from thermal plants while satisfying the generation constrained such as generation limits, prohibited operating zones, and ramp-rate limitation of generators. The cost and emission coefficient of objective function a , b , and c and α , β , and γ for this system is outlined in [72]. Compared with earlier case this scenario is more complex as it has two conflicting objectives and necessitates a trade off between the functions presented in equations (2) and (6). The power demand for this scenario is 10500 MW and total 40 generators are participating in ring topology.

FIGURE 10 illustrates the progression of an objective function throughout iterations. All system restrictions, including VLE and restricted operating zones, are included into the optimization problem. Two hundred distinct runs were conducted, revealing that the proposed Whale optimization scheme yields notable results by decreasing the chaotic cost and emission functions while adhering to all system restrictions. The convergence of the cost and emission functions is illustrated graphically in FIGURE 11. The suggested strategy rapidly converges to the global optimum, demonstrating its effectiveness in identifying the global optimal

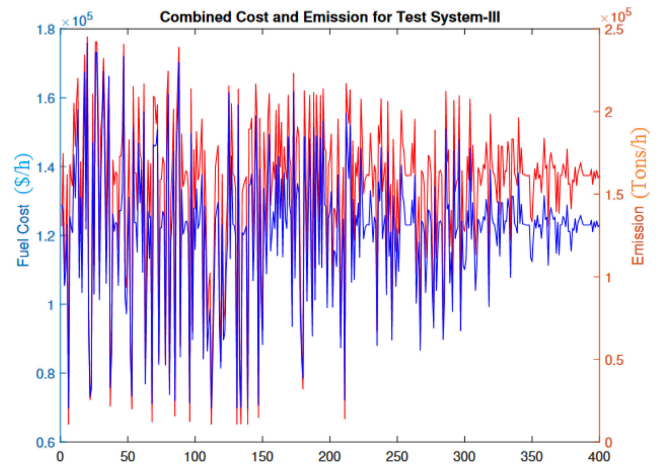


FIGURE 11. Convergence Cost and Emission Function [72].

solution in less iterations, while simultaneously exhibiting reduced computational time, as seen in FIGURE 10.

VII. CHALLENGES AND FUTURE WORKS

OPF has many benefits for the operation and planning of power systems, such as enhancing system efficiency, decreasing operation costs, and managing the integration of RES. However, OPF also has several issues in its calculations, such as complexity problems, how to deal with stochastic factors, and how to maintain system stability in different scenarios. Based on the various research articles reviewed in the earlier sections on OPF techniques and methodologies, it can be deduced that there are still many opportunities to further improve them and make new advances. Below is a summarized list of challenges and recommendations that will enable researchers to harness advanced technology to enhance power system operations using OPF:

- Further research could emphasize identifying new and improved approaches for addressing uncertainty in the OPF problem. This includes using tools such as probabilistic forecasting, scenario planning, and robust optimization strategies [210]. These approaches could immensely improve the OPF solutions' reliability and accuracy, especially under varying conditions, and offer better decision-making frameworks in times of volatility, thereby guaranteeing much stronger and more optimal power system operations.
- Most studies to date have focused on centralized OPF, where a single centralized entity manages the optimization. Nonetheless, due to the increasing incorporation of DERs, there is a noticeable demand for distributed OPF approaches. These methods would help integrate the DERs most efficiently while ensuring the stability of the power system as well as its reliability.
- It might be possible to use more research to create algorithms that combine ideas about dependability, operation, and security into a single framework that determines OPF. By addressing all of these aspects

simultaneously, future algorithms could offer better and more complete solutions to how the power system should perform optimally and securely under so many different scenarios.

- There still exist small gaps of knowledge in research fields that are related to the use of machine learning in the management and planning of power systems. Further research could look into the benefits of deep learning for OPF. This could be used for machine learning to help OPF solvers get started or to find key input variables for training models [211]. Researchers could keep looking into how safe trained models are, how to better tell the difference between active and inactive constraints, and how to use machine learning to guess binary variables related to unit commitment and expansion planning.
- Traditional OPF studies typically do not include multiple contingencies, despite their potential to seriously affect system stability and reliability. Filling this gap offers a potential for future research, which could enhance the resilience of power systems.
- It is very hard to do real-time OPF that includes BSSs, especially when it is thought of as a dynamic mixed-integer nonlinear programming (MINLP) problem with uncertainty [212]. The need to manage the fluctuating nature of renewable energy sources and the variability in demand heightens this complexity.
- Issues of battery and power electronics' longevity and reliability in real-time OPF are still open. Guaranteeing that these devices will be capable of withstanding operational challenges for an extended period without straining the system's efficiency and stability is the key to the future of controlling power systems. Solving these issues implies further developments not only in the theoretical approaches based on mathematical modeling but also in the practical solutions as well.

VIII. CONCLUSION

In recent years, OPF has become an increasingly important tool for power system operation and planning, particularly with the growing integration of renewable energy sources, DERs, and ESS. While OPF is key to optimizing power distribution, advanced optimization techniques are essential for handling the increasing complexity of modern power grids. This paper offers a comprehensive review of OPF theoretical concepts, mathematical formulation, and current state-of-the-art literature on OPF methodologies encompassing conventional and advanced approaches. This work has reviewed various optimization techniques, including mathematical, artificial intelligence, metaheuristic, machine learning, and hybrid methods, highlighting the developments and challenges of each method. Furthermore, the discussion includes the challenges associated with OPF, such as computational complexity, uncertainty integration, and real-time implementation. The authors of this study identify key gaps and future research directions to guide the development of more robust and efficient OPF solutions. The paper

provides important insights, a deeper understanding, and future research directions for those interested in OPF.

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